Inverse Method for Interface Problems

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(Received 22 February 1993)

We propose an inverse method to extract effective couplings and the renormalization group flow for macroscopic dynamical systems. We applied it to discrete surface growth models in the Kardar-Parisi-Zhang universality class in $1+1$ dimensions and obtain the first measurement of a universal coupling constant. It may also be applicable to analyze experimental data such as immiscible fluid displacement experiments.

PACS numbers: 64.60.Ht, 05.40.+j, 05.70.Ln, 64.60.Ak

Continuum evolution equations contribute significantly to our understanding of growth of random self-similar surfaces [1]. Examples include the Edwards-Wilkinson (EW) equation describing sedimentation [2] and the Kardar-Parisi-Zhang (KPZ) equation describing vapor deposition and directed polymers [3]. The height-height correlation function of these surfaces in the steady state follows the scaling form $\langle|h(x,t)-h(0,t')|^2\rangle \sim |x|^{2\alpha}f[|t-t'|/|x|^z]$. The roughness exponent α and the dynamical exponent z characterize the universality classes.

Interfaces following such scaling have been observed, respectively, in several recent experiments on fluid displacement in porous media [4], bacteria colony expansion [5], and fire fronts in paper [6]. The measured values of α scatter around 0.6–0.9. Much effort has been concentrated on theoretical modeling, including growth models with power-law noise [7], long-range correlated noise [8], and models where local pinning is important [9]. Most of these possible explanations contain free parameters so that the predicted values of α, or effective α, cover wide and overlapping ranges. This leads to severe problems in interpreting the experimental data: the exponents alone give very little insight into the fundamental growth process. Recently, Horváth *et al.* computed the tail of the noise distribution and correlation in a fluid displacement experiment and indicated the existence of power-law noise [10]. Thorough analysis of the experiments along such lines is evidently necessary for better understanding of the dynamics. The deficiency of scaling exponents in identifying universality classes is also evident in the investigations of models with long crossovers. Examples are the Toom model [11] and growth with power-law noise [12] or correlated noise [13] where logarithmic corrections to scaling have been suggested.

Here we propose an inverse method to find essentially complete information about growth processes. It closely resembles the spirit of Ma's Monte Carlo renormalization group, which computes numerically the effective block spin Hamiltonian from equilibrium spin configurations [14]. Our method can be considered as the dynamical and continuum counterpart and is applicable for both equilibrium or nonequilibrium systems. We calculate directly the evolution equation for a given surface from simulation or possibly experimental data. The renormalization group flow of the equation can be obtained with *all* its parameters computed. The method is as follows: A time sequence of snapshots of a surface captures its dynamical properties up to the corresponding resolution. It is usually not difficult to guess a general candidate equation for the evolution containing all the possible terms consistent with symmetry. Our task is to discriminate among them. Given a snapshot of the surface, realizations of the equation with different coefficients give different predictions about the next snapshot. The correct equation is the one which, on the average, gives the best predictions over a large number of trials. It can thus be obtained by minimizing the errors of the predictions. Any terms with zero coefficients are proved to be absent.

We illustrate the method by working through applications to growth models in the KPZ universality class in $1+1$ dimensions. Though these examples have been extensively studied previously, this novel technique is powerful enough to yield entirely new and important results as well as reproduce the known ones. Extension to other less understood cases is straightforward. In the present case, the evolution of the surface is known to be given by the KPZ equation [3]

$$\frac{\partial h}{\partial t} = c + \nu\nabla^2 h + \frac{\lambda}{2}(\nabla h)^2 + \eta(x,t), \tag{1}$$

where the noise η has a Gaussian distribution and mean 0 and a correlator given by

$$\langle\eta(x,t)\eta(x',t')\rangle = 2D\delta(x-x')\delta(t-t'). \tag{2}$$

In addition, there is a lower wavelength cutoff, or spatial resolution, l. From dynamical renormalization group (DRG) calculations, for a given surface, ν and D depend on the resolution l of observation and, asymptotically for large l, we have [3,15]

$$\nu(l) \sim l^{\alpha}, \qquad D(l) \sim l^{\alpha}. \tag{3}$$

The parameters λ and $D(l)/\nu(l)$ are independent of l due to symmetry [8] and a fluctuation dissipation theorem [15], respectively. Both of them have been measured numerically [16–18] and, for some particular models, can be obtained analytically [17,19].

0031-9007/93/71(4)/561(4)$06.00

We take the candidate evolution equation to be in a generalized form:

$$\frac{\partial h}{\partial t} = c + \nu\partial^2 h + \frac{\lambda}{2}(\partial h)^2 + a_4\partial^4 h + a_5\partial^2(\partial h)^2 + a_6\partial h\partial^3 h + a_7\partial^6 h + \cdots + \eta, \tag{4}$$

where all the differentiations on the right are with respect to x. Discretizing time, we have

$$\frac{\Delta h(x,t)}{\Delta t} \simeq \mathbf{a}\cdot\mathbf{H}(x,t) + \eta(x,t), \tag{5}$$

where $\mathbf{a} = (c, \nu, \frac{\lambda}{2}, a_4, ...)$ and $\mathbf{H}(x,t) = (1, \partial^2 h, (\partial h)^2, \partial^4 h, ...)$ are vectors. We find the parameter vector $\mathbf{a}$ so that the equation best describes the snapshots. To measure $\mathbf{H}(x,t)$ numerically, we first simulate a given discrete model to obtain an initial discrete surface $h_d(x,t)$. We coarse grain it by truncating the Fourier components of wavelengths smaller than l. From the coarse-grained surface, we get $\mathbf{H}(x,t)$ which thus depends on l. The differentiations and multiplications involved are done in the Fourier and the real spaces, respectively. An additional coarse graining is then carried out to maintain the cutoff which is shifted by the nonlinearities. We obtain m realizations of $h_d(x, t+\Delta t)$ by further growing the *same* surface $h_d(x,t)$ m times. Averaging gives the mean discrete profile at time $t+\Delta t$ from which we obtain the coarse-grained mean surface advance $\langle\Delta h(x,t)\rangle$, which depends on both the spatial and temporal resolutions l and Δt. For large m, Eq. (5) averages to

$$\frac{\langle\Delta h(x,t)\rangle}{\Delta t} \simeq \mathbf{a}\cdot\mathbf{H}(x,t). \tag{6}$$

The fitting can be done most conveniently with discrete data sets. We look at N discrete points (x_i, t) each separated spatially by l and denote $\mathbf{H}_i = \mathbf{H}(x_i, t)$ and $\langle\Delta h_i\rangle = \langle\Delta h(x_i,t)\rangle$. To enhance the statistics, the data set can be taken from M realizations of the pair $H(x,t)$ and $\langle\Delta h(x,t)\rangle$ generated from the corresponding M realizations of $h_d(x,t)$. Using a least square fitting procedure, the mean square deviation of Eq. (6) from equality is

$$\mathcal{D} = \frac{1}{N}\sum_{i=1}^{N}\left(\frac{\langle\Delta h_i\rangle}{\Delta t} - \mathbf{a}\cdot\mathbf{H}_i\right)^2. \tag{7}$$

Minimizing $\mathcal{D}$ with respect to $\mathbf{a}$ gives the main equation of our inverse method, from which $\mathbf{a}$ can be found,

$$\mathbf{A}\mathbf{a} = \mathbf{b}, \tag{8}$$

where the matrix $\mathbf{A}$ and the vector $\mathbf{b}$ are defined by

$$\mathbf{A} = \frac{1}{N}\sum_{i=1}^{N}\mathbf{H}_i\otimes\mathbf{H}_i, \qquad \mathbf{b} = \frac{1}{N}\sum_{i=1}^{N}\frac{\langle\Delta h_i\rangle}{\Delta t}\mathbf{H}_i. \tag{9}$$

The symbol $\otimes$ denotes the vector outer product. Straightforward algebra gives the noise correlator as

$$D = \Delta t\; l\; \mathcal{D}'/2, \tag{10}$$

where $\mathcal{D}'$ is defined like $\mathcal{D}$ in Eq. (7) with $\langle\Delta h_i\rangle$ replaced by an unaveraged Δh_i. We have assumed the δ correlations in Eq. (2). We have now obtained expressions for all the parameters in the continuum formulation.

One can easily see that the same value of $\mathbf{a}$ would be obtained if we were to minimize $\mathcal{D}'$ instead of $\mathcal{D}$. Therefore, m is not necessarily large and can even be 1, which means no averaging of surface advance. The method might then be applied to analyze experimental surfaces. For simulations, the averaging improves the computational speed.

We tested the method by simulating a simple discrete model in the universality class of the Edwards-Wilkinson equation [2], which is the linear version of the KPZ equation with $\lambda = 0$. Except for small l where there are errors due to discretizing time in Eq. (5), the values of ν and D obtained using Eqs. (8) and (10) are independent of l and Δt and are in good agreement with the exactly known values. In particular, λ is measured to be 0 with excellent precision.

We then applied the method to four discrete models in the KPZ universality class. Two of them are single step models [20] in which the nearest-neighbor height differences of the surface are constrained to be ± 1. A particle of height two can be added to the aggregate only if the resulting step heights obey the constraint. The first model (SS I) is the original single step model, where there is one growth attempt at a random site per site per unit time [20]. The other one (SS II) adopts sublattice updating. During each time step, growth is attempted independently with 50% probability at all odd sites and then at all even sites [19]. We also simulated two variants of the ballistic deposition model with sublattice updating (BD I and BD II) with growth rules given by $h(x,t+1) = \max\{h(x,t)+\eta(x,t), h(x-1,t), h(x+1,t)\}$ and $h(x,t+1) = \frac{1}{2}[\max\{h(x,t), h(x+1,t), h(x-1,t)\} + h(x,t)] + \eta(x,t)$ [12], respectively, where the η's are independent uniform random deviates.

For each model, we simulated growth on substrates of size 32 768 with periodic boundary conditions. We focus only on surfaces in the steady state. We generated $M = 1000$ realizations of the fully developed surface $h_d(x,t)$. For the SS models, the probability distribution of the steady surface is known [19,20] and the surfaces can be generated directly. The BD surfaces are generated by properly rescaling SS I surfaces followed by further growth for 2000 time steps with the BD rules. For every $h_d(x,t)$, the number of realizations of $h_d(x,t+\Delta t)$ is typically $m = 20$. The fittings are done with the seven terms listed in Eq. (4) using Eqs. (8) and (10) for various values of l and Δt.

The measured model parameters for the SS I model are shown in Fig. 1. The statistical errors are small except noticeably for ν at large l where the magnitude of this term is small compared with the others. For $l^z \lesssim \Delta t$, where $z = \frac{3}{2}$ in 1 + 1 dimensions, all the parameters

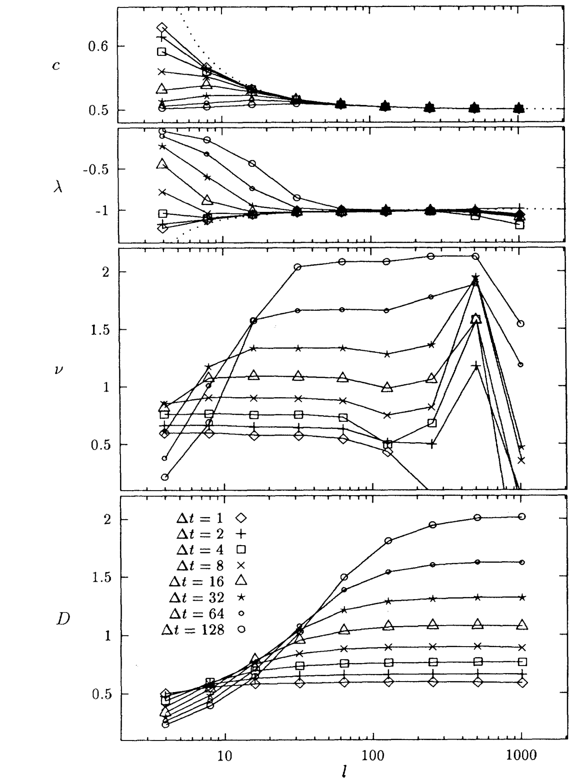

FIG. 1. Measured continuum parameters c, λ, ν, and D as functions of the resolutions l and Δt for the SS I model.

are affected by the finite time difference approximation. The coefficients of the higher order terms (not shown) admit much larger statistical error and their values are dominated by the finite difference effects. From Fig. 1, both c and λ converge to the expected exact values with the "finite size corrections" $\frac{1}{2}[1-(l-1)^{-1}]^{-1}$ and $-[1-(l-1)^{-1}]^{-1}$, respectively (dotted lines) [21]. The parameters ν and D admit renormalization and are much more interesting. Both ν and D converge as l increases and become functions of Δt only. Table I lists the numerical values of ν, D and D/ν at $\Delta t = 128$ for large l together with λ. Bracketed values are exact results [17,19]. The errors are estimated to be around 2% to 5%. The values of D/ν measured agree reasonably with the exact results but are systematically smaller because D, which is related to the error in the fitting, is underestimated due to the finite sample size. The agreement supports the δ correlation of the renormalized noise assumed in deriving Eq. (10). Short-range spatial correlations do exist but their effects diminish with increasing l, explaining the slower convergence of D compared to ν as seen in Fig. 1. We do not yet fully understand why temporal correlations seem to be absent.

Since the DRG calculations look at the long time behavior [15] while we have finite observation time Δt, the

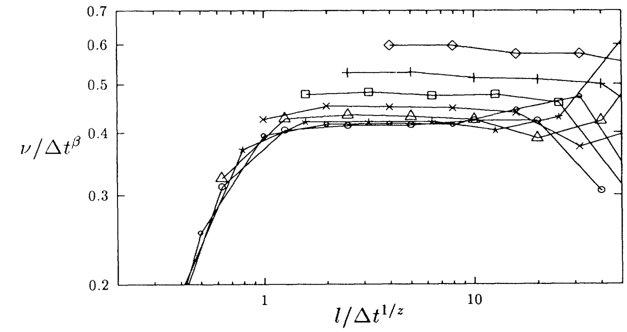

FIG. 2. Data collapse of the values of ν from Fig. 1.

asymptotic form in Eq. (3) is generalized to

$$\nu(l,\Delta t) = l^{\alpha} f_1(\Delta t/l^z) = \Delta t^{\beta} f_2(\Delta t/l^z) \tag{11}$$

and similarly for D, where the early time exponent $\beta = \frac{\alpha}{z} = \frac{1}{3}$. The convergence of ν and D, which manifests as $\lim_{y\to 0} f_2(y) \to$const, corresponds to the fact that only Δt controls the extent of the renormalization for sufficiently large l. This will be explained further later. With increasing Δt, the asymptotic scaling form (11) is expected and is demonstrated through the reasonable data collapse for ν at $\Delta t \gtrsim 32$ for the SS I model in Fig. 2. The result for D is similar. The success of the data collapse demonstrates that our method yields the correct exponents. By examining curves similar to Fig. 2 for different exponents, we can show that we must have $\beta = 0.33 \pm 0.01$ and $z = 1.50 \pm 0.04$.

From DRG calculations [3,15], asymptotic scaling is approached when a dimensionless coupling constant $\overline{\lambda}^2(\Lambda) = \lambda^2 D/\nu^3 \Lambda$ flows to a finite universal fixed point upon renormalization in $1+1$ dimensions, where $\Lambda = \pi/l$ is the upper momentum cutoff. Now the extent of renormalization is controlled by Δt and Λ should be replaced by an effective cutoff $c(\lambda^2 D/\nu)^{-1/3}\Delta t^{-1/z}$, where c is a universal constant and the remaining proportionality constants are from dimensional analysis. This leads to a redefined coupling constant $\overline{\lambda}^2(\Delta t) = [(\lambda^4 D^2/\nu^5)\Delta t]^{2/3}$. The numerical values are listed in Table I. The model independence is clearly demonstrated, though additional errors exist because the coupling constants are close to but not fully converged at $\Delta t = 128$. We obtained

TABLE I. Numerical values of λ, ν, and D for $\Delta t = 128$, which give D/ν and $\overline{\lambda}$. Bracketed values are exact results.

Model	λ	$\nu/\Delta t^{1/3}$	$D/\Delta t^{1/3}$	D/ν	$\overline{\lambda}^2$
SS I	-1.00 (-1.00)	0.413	0.399	0.966 (1.000)	5.66
SS II	-0.712 (-0.707)	0.254	0.236	0.929 (0.971)	5.68
BD I	1.86	0.253	0.0325	0.128	5.30
BD II	1.64	0.265	0.048	0.181	5.45

$\overline{\lambda}^2(\infty) = 5.5 \pm 0.2$, which is the first numerical estimate of the universal fixed point.

To understand the qualitatively different behavior due to adopting a finite observation time, we examine the solution of the KPZ equation expressed in momentum space [3],

$$h(k,t+\Delta t) = G_0(k,\Delta t)h(k,t) + \int_t^{t+\Delta t} d\tau G_0(k,t+\Delta t-\tau)\left(\eta(k,\tau) - \frac{\lambda}{2}\int_{-\Lambda}^{\Lambda}\frac{dq}{2\pi}q(k-q)h(q,\tau)h(k-q,\tau)\right), \quad (12)$$

where the propagator is $G_0(k,\Delta t) = \exp(-\nu k^2 \Delta t)$. Corrections to the propagator come from iterating Eq. (12) and averaging over $\eta(q,\tau)$ and $h(q,t)$ in a short wavelength shell. For $\Delta t \to 0$, the integrals can be approximated by multiplications and the nth perturbative term is of order $(\lambda \Delta t)^n$, leading to a renormalization of ν of order $\lambda^n \Delta t^{n-1}$. If the correction from the first term vanishes, the overall correction becomes $\mathcal{O}(\Delta t) \to 0$ as $\Delta t \to 0$. For fixed Δt, the short time limit corresponds to large l such that $\Delta t \ll l^z$. Therefore for sufficiently large l, further increase in l, by integrating a short wavelength shell, leads to no more correction and the convergence of ν is justified. Intuitively, Δt controls the extent of the renormalization by dictating how short the wavelength of the modes should be to evolve fast enough to lead to renormalization.

We now examine whether the correction due to first perturbative term vanishes. The effective propagator is defined by $h(k,t+\Delta t) = G(k,\Delta t)h(k,t)$, which gives $\langle h(k,t)h(k,t+\Delta t)\rangle = G(k,\Delta t)\langle h^2(k,t)\rangle$ [22]. The contribution of the first perturbative term to the left-hand side of the equation contains the factor $F(k,q,t) = \langle h(k,t)h(k-q,t)h(q,t)\rangle$ in the integrand. For 1+1 dimensions, the fluctuation dissipation theorem implies that all the $h(k,t)$'s are independent in the steady state [15] so that the factor $F \equiv 0$ and the result follows. However, in general, the $h(k,t)$'s are correlated and $\nu(l,\Delta t)$ and $D(l,\Delta t)$ do not necessarily converge as $l \to \infty$.

In conclusion, we introduced an inverse method for thorough analysis of macroscopic dynamics systems. Applications to a class of growth models yield the first computation of *all* the parameters in the corresponding KPZ equation. This is the most direct numerical verification of the applicability of the equation. The finiteness of the universal coupling constant is first proved numerically with its value computed. In 2+1 dimensions, the dynamical renormalization group technique has failed to access the strong coupling fixed point. Our inverse method is a unique and very promising technique to study the nature of this fixed point which is of central importance to possible analytical approaches. The method should have wide applicability including analyzing experiments surfaces.

We thank D. S. Graff, F. Guinea, D. A. Kessler, D. Ko, O. Pla and R. B. Stinchcombe for interesting discussions.

This work is supported by NSF Grant No. DMR91-17249.

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